

Prakhar Singh

League City, TX • prakhar20117@gmail.com • [Website](#) • [GitHub](#)

EDUCATION

Clear Springs High School

Class of 2029

Weighted GPA: 5.96 • Unweighted GPA: 4.0

ACTIVITIES

TSA — Engineering Design

Oct 2025 – Feb 2026

Team Member

- Collaborated with 3 peers to design and prototype a pocket-sized telescope for mass production
- Developed CAD models, project portfolio, and competition displays over a 5-month build cycle
- Qualified for and competed at the district level

SEFH — Gut Microbiome Research

2023 – 2024

Independent Researcher

- Designed and conducted experiment testing effects of sucrose, aspartame, fructose, and other sweeteners on Lactobacillus growth
- Found aspartame to be particularly harmful to gut bacteria compared to natural sugars
- Placed 1st at district; earned Honorable Mention at the Science and Engineering Fair of Houston (SEFH)

Boy Scouts of America

2023 – Present

Patrol Leader & SPL Scribe

- Led patrol as Patrol Leader; earned 2nd place in troop-wide cooking competition
- Serve as scribe under the Senior Patrol Leader for two terms (1 year), recording minutes, tracking attendance, and managing troop communications
- Working toward Eagle Scout rank with a self-directed service project

SKILLS

CAD Modeling

Designed mechanical models including an automaton, LEGO truck, scale theme park, and robot drivetrain components.

LLM-Assisted Development

Built this personal website from scratch using HTML, CSS, and JavaScript with AI collaboration. Skilled in prompt engineering, code review, and iterative debugging with LLMs.

JavaScript

Built a functioning 2D runner game inspired by Flappy Bird; can create encoders, decryptors, and image filters adaptable as apps.

Collaborative Problem Solving

Adaptable across independent research (SEFH) and team engineering (TSA). Confident in iterative design, prototyping, and presenting technical work.